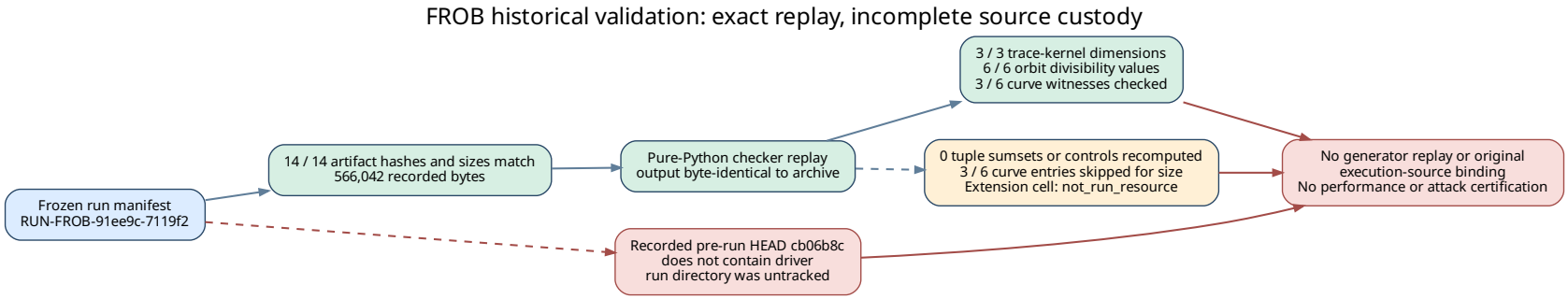
Historical FROB checker replay and source custody

Validation audit, 22 October 2026

Scope. This checks frozen run `RUN-FROB-91ee9c-7119f2` at source tree `e83bb2f14414c9669360816aeee5876dfcfd4384`. It is an integrity audit, not a new FROB measurement or a research verdict. The run directory was copied before executing its pure-Python checker; frozen bytes were not edited.

Check	Result	Limit
Archived artifact hashes and sizes	14/14 match; 566,042 bytes	Storage integrity only
Checker output	Byte-identical replay	Same archived checker source
Trace-kernel dimension; orbit divisibility	3/3; 6/6	Recorded core cells
Curve point/order/Frobenius witnesses	3/6 checked	Other three skipped for size
Tuple sumsets and C2/C3/C4 arms	0 recomputed	Checker source confirms this limit
Extension cell	`not_run_resource`	540-second historical checkpoint



Source custody. The manifest records pre-run HEAD `cb06b8c20c2d6559041c45c4a4b29dd7f132fdf5` and says the run directory was untracked at execution start. The driver is absent from that commit. Current-schema `code.commit`, `code.command`, and `environment` fields are absent, although separate command and environment artifacts are hash-pinned. Their presence does not bind the executed driver to the recorded HEAD.

Boundary and next check. The archived checker's `tuple_sum_reverifications` label overstates its code: its three `ok` entries check curve witnesses, not tuple sums or control arms. The replay does not rerun the Sage generator or any performance experiment. Recover a contemporaneous execution snapshot and exact command if available; otherwise run a fresh source-bound replay against the frozen specification and controls. The Coordinator must adjudicate the original record without assigning the pre-run HEAD as a false execution-source commit. Machine receipt: `docs/reports/2026-10-22-frob-provenance-audit/audit.json`.